

National Institute of Technology, Kurukshetra
Department of Electronics and Communication Engineering

HPC-212 Data Communication & Networking Max. Marks: 15 Time: 50 Minutes

Note: All Questions are compulsory

- Q1. A data packet is sent from a web server to a client. Identify the layers involved in this communication in OSI model and describe their roles. (3)
- Q2. A digital signal has a bit rate of 10 Mbps. Calculate the time it takes to send a file of size 2 GB over this signal. (3)
- Q3. A communication system has a bandwidth of 4 kHz and a signal-to-noise ratio (SNR) of 20 dB. Calculate both the Nyquist bit rate and the Shannon capacity for this system. (4)
- Q4. Given a digital signal represented by the sequence 110100110, draw the line coding representation for MLT-3 (Multi-transmission) and bipolar AMI (Alternate Mark Inversion) schemes. (3)
- Q5. Describe three different types of transmission media used in data communication. (2)